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|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Name of the Student</b> | <b>P.Hithaishi</b>                                                                                              |
| <b>Reg. No</b>             | <b>192425196</b>                                                                                                |
| <b>GitHub Link</b>         | <a href="https://github.com/hithu14/MLA0403-Deep-Learning">https://github.com/hithu14/MLA0403-Deep-Learning</a> |

## SIMATS ENGINEERING

### CO2 - ASSESSMENT TOOL 2

#### Scenario-Based Questions

**COURSE NAME: Deep Learning**

**COURSE CODE: MLA04**

#### **COURSE OUTCOME COVERED**

**CO 2:**Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

**CO Weightage : 15%**

**Assessment Tool Weightage: 40%**

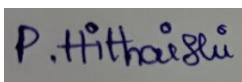
**Duration: 90 minutes**

**Marks: 25**

#### **Code of Conduct:**

I P. Hithaishi (192425196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



**Signature of the student:**

**Name of the student: P. Hithaishi**

## Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

### 1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

#### Problem

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

#### Recommended Technique

##### Autoencoder (Unsupervised Neural Network)

#### Justification

Hospitals generate thousands of ECG signals every day, but most of them are not labeled by medical experts because manual labeling is time-consuming and expensive. Autoencoders are ideal for such situations because they learn the underlying structure of normal ECG signals without requiring labeled data.

After training, the model reconstructs normal ECG signals with very small error. However, abnormal ECG signals differ from the learned pattern and therefore produce a much larger reconstruction error. This difference can be used to detect heart abnormalities automatically.

#### Workflow

1. Collect unlabeled ECG recordings.
2. Remove noise and normalize the ECG signals.
3. Train the autoencoder using normal ECG patterns.
4. The encoder compresses the ECG into a low-dimensional feature vector.
5. The decoder reconstructs the ECG from the compressed representation.
6. Compute the reconstruction error.
7. Compare the error with a threshold value.

8. If the error is high, classify the ECG as abnormal and notify the doctor.

### **Advantages**

- No labeled data is required.
- Detects unknown or rare heart abnormalities.
- Learns meaningful ECG features automatically.
- Supports continuous patient monitoring.
- Reduces diagnosis time and assists cardiologists.

### **Applications**

- Arrhythmia detection
- Heart disease screening
- ICU patient monitoring
- Wearable health devices

### **Conclusion**

Autoencoders provide an efficient and reliable method for detecting abnormal ECG signals by learning normal heart activity and identifying unusual patterns through reconstruction error.

## 2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

### **Problem**

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation.

### **Solution Using Autoencoder**

Retail companies collect customer information such as age, purchase frequency, spending amount, product categories, payment methods, discounts used, and shopping preferences. Since the dataset contains many features, it becomes difficult and computationally expensive to analyze.

An autoencoder compresses this high-dimensional data into a smaller latent representation while preserving the most important information.

### **Working Process**

1. Collect customer purchase records.
2. Normalize and preprocess the dataset.
3. Feed the data into the autoencoder.
4. The encoder learns compressed customer representations.
5. The decoder reconstructs the original purchase data.
6. Minimize reconstruction error during training.
7. Extract latent features from the bottleneck layer.
8. Apply clustering algorithms such as K-Means to the compressed features.

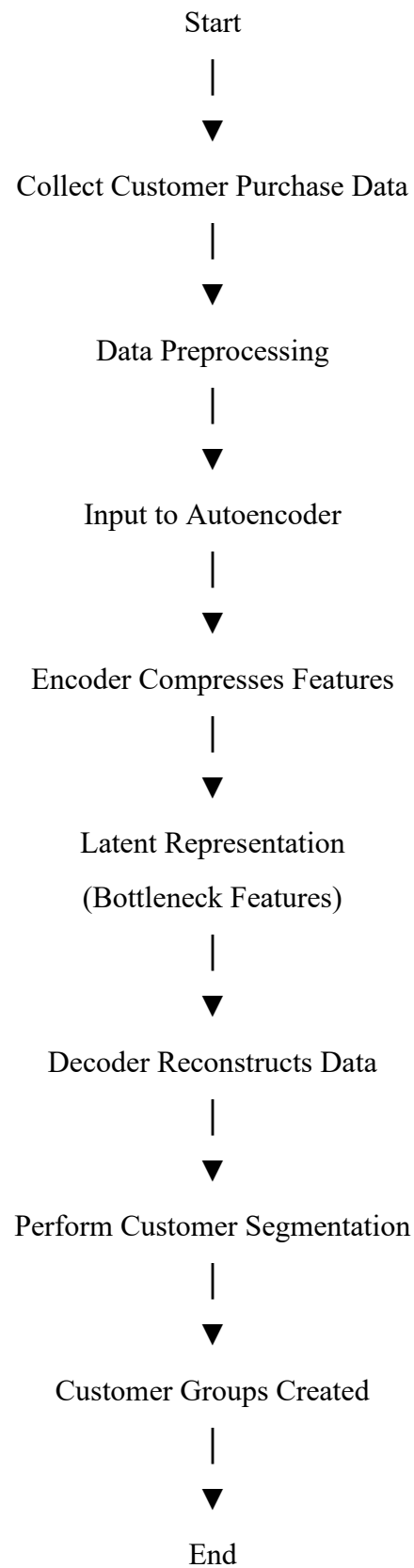
### **Benefits**

- Removes redundant and noisy features.
- Preserves essential customer behavior.
- Improves clustering quality.
- Reduces computational complexity.
- Enables faster customer analysis.

### **Applications**

- Customer segmentation

- Personalized product recommendations
- Targeted marketing campaigns



**Conclusion**

Autoencoders perform nonlinear dimensionality reduction, making customer segmentation more accurate and efficient than traditional feature reduction methods.

### 3. **Width and Depth of Neural Networks, Activation Functions**

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

#### **Problem**

A facial recognition system performs poorly when trained using a shallow neural network.

#### **Recommended Architectural Changes**

A shallow neural network cannot effectively learn complex facial structures such as eyes, nose, lips, facial expressions, illumination changes, and different viewing angles.

Increasing the depth and using suitable activation functions allows the network to learn hierarchical image features.

#### **Recommended Changes**

- Increase the number of hidden layers (greater depth).
- Increase the number of neurons where necessary (greater width).
- Use **ReLU** activation in hidden layers.
- Use **Softmax** activation in the output layer.
- Apply Batch Normalization.
- Use Dropout to reduce overfitting.
- Train with a larger facial image dataset.

#### **Role of Depth**

- Early layers learn edges.
- Middle layers learn facial components.
- Deep layers learn complete face representations.

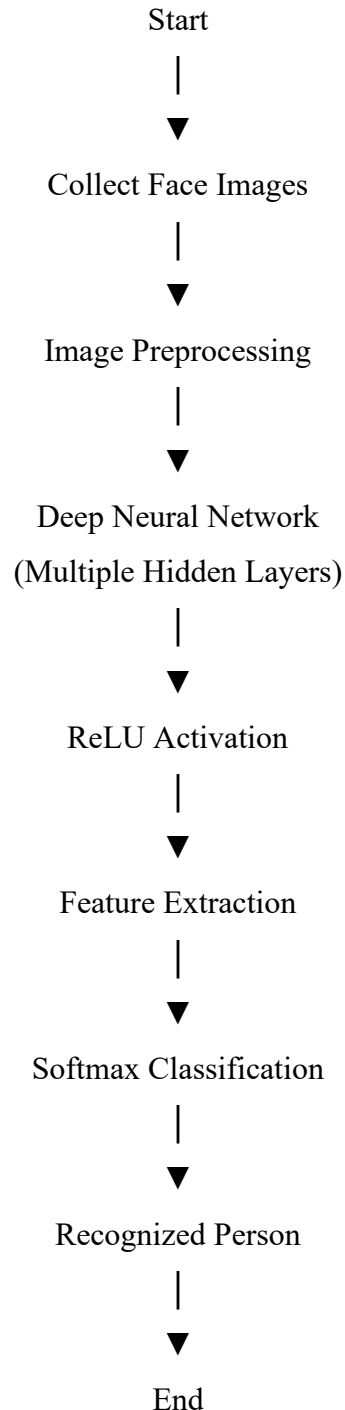
#### **Role of Width**

- Learns more feature variations.
- Improves representation capacity.
- Captures multiple facial characteristics simultaneously.

#### **Advantages**

- Better feature extraction.

- Faster convergence.
- Reduced vanishing gradient problem.
- Improved recognition accuracy.
- Better performance under different lighting and facial expressions.



### **Conclusion**

A deeper neural network combined with ReLU activation significantly improves facial recognition accuracy by learning complex hierarchical facial features.

#### **4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**

A streaming service wants to recommend movies to users based on their viewing history without using labelled data. Explain how RBMs can be applied to build the recommendation engine.

##### **Problem**

A streaming service wants to recommend movies to users based on their viewing history without using labelled data.

##### **Solution Using RBM**

A Restricted Boltzmann Machine is a probabilistic neural network that learns hidden relationships between users and movies without supervision.

It consists of:

- **Visible Layer:** User ratings or watch history.
- **Hidden Layer:** Latent preferences such as comedy, action, thriller, romance, or science fiction.

##### **Working Process**

1. Collect user viewing history.
2. Convert movie ratings into a user-item matrix.
3. Initialize RBM weights randomly.
4. Train using Contrastive Divergence.
5. Learn hidden user preferences.
6. Predict missing movie ratings.
7. Recommend movies with the highest predicted ratings.

##### **Advantages**

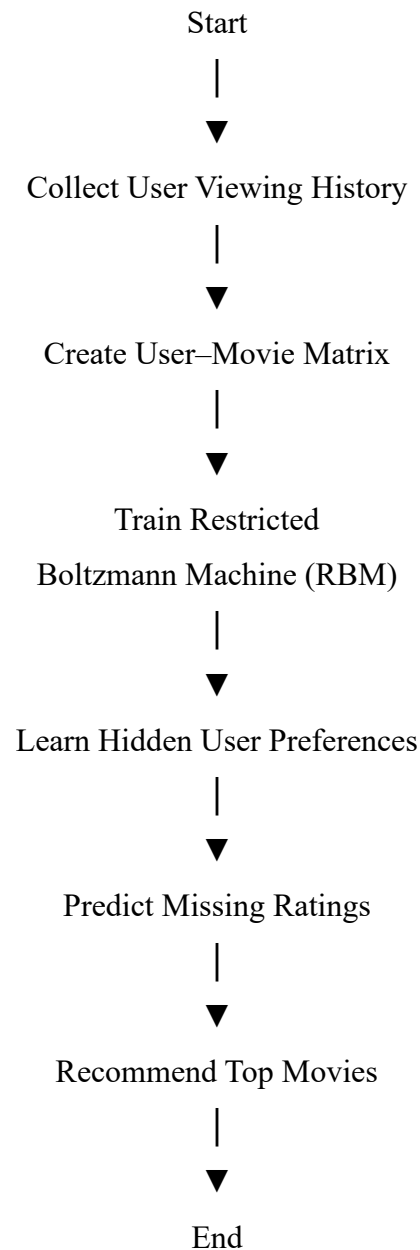
- Learns hidden viewing preferences.
- Handles sparse rating matrices.
- Requires no labeled recommendation data.
- Produces personalized recommendations.
- Discovers relationships between users and movies automatically.

##### **Applications**

- Netflix recommendations



- Amazon product recommendations
- Spotify music recommendations
- Online shopping suggestions



### **Conclusion**

RBMs are powerful unsupervised models that capture hidden user interests and generate accurate personalized recommendations.

## 5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

### Problem

A manufacturing company wants to detect defective products from sensor readings collected every second.

Proposed Deep Learning Solution

Use an Autoencoder-based Anomaly Detection System.

The system learns normal machine behavior from sensor data. When defective products or machine faults occur, reconstruction error increases, allowing automatic defect detection.

### Workflow

1. Collect sensor readings continuously.
2. Remove missing values and normalize data.
3. Train the autoencoder using normal production data.
4. The encoder extracts compact feature representations.
5. The decoder reconstructs sensor readings.
6. Calculate reconstruction error.
7. Compare error with a threshold.
8. Generate alerts when abnormal readings are detected.

### Representation Learning

The encoder automatically learns important sensor patterns such as:

- Temperature trends
- Pressure variations
- Machine vibration
- Motor speed
- Production quality indicators

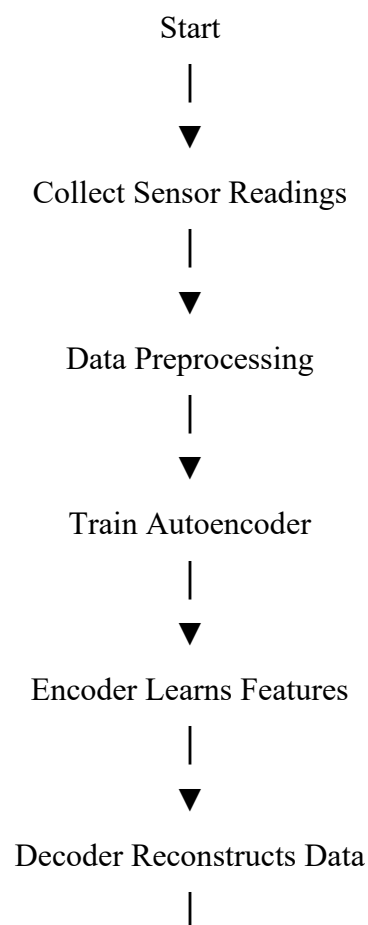
These learned representations improve anomaly detection without manual feature engineering.

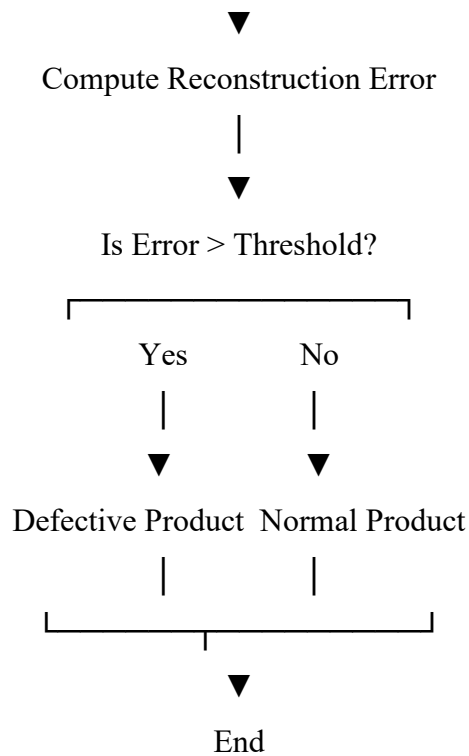
### **Advantages**

- Detects defects in real time.
- Requires only normal training data.
- Reduces production losses.
- Supports predictive maintenance.
- Improves manufacturing quality.
- Reduces manual inspection cost.
- Increases production efficiency.

### **Applications**

- Smart manufacturing
- Industrial IoT systems
- Predictive maintenance
- Quality control
- Fault detection in production lines





### Conclusion

Representation learning with autoencoders enables fast and accurate detection of defective products by learning normal sensor behaviour and identifying abnormal patterns automatically. It improves product quality, reduces downtime, and supports intelligent manufacturing systems.

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                  | <b>Level 3 – Proficient</b>                       | <b>Level 2 – Developing</b>                                 | <b>Level 1 – Beginning</b>              |
|---------------------------|------------------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%              | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%              | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%              | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%              | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |
| Clarity                   | 10%              | Concept map is clear, neat, visually structured, and easy to interpret      | Generally clear with minor readability issues     | Clarity is reduced; difficult to interpret in parts         | Poorly presented and hard to understand |